

Viromics

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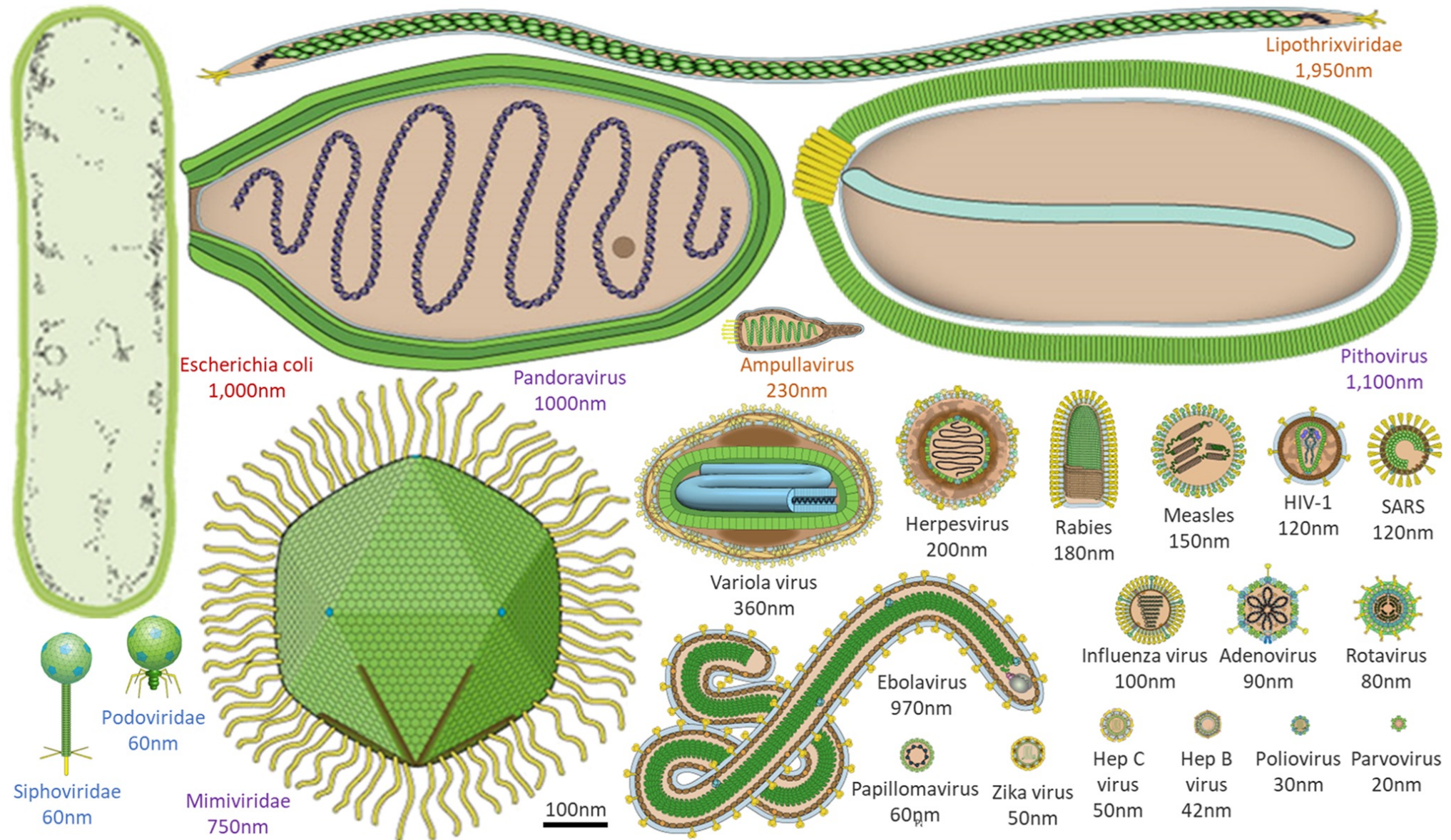
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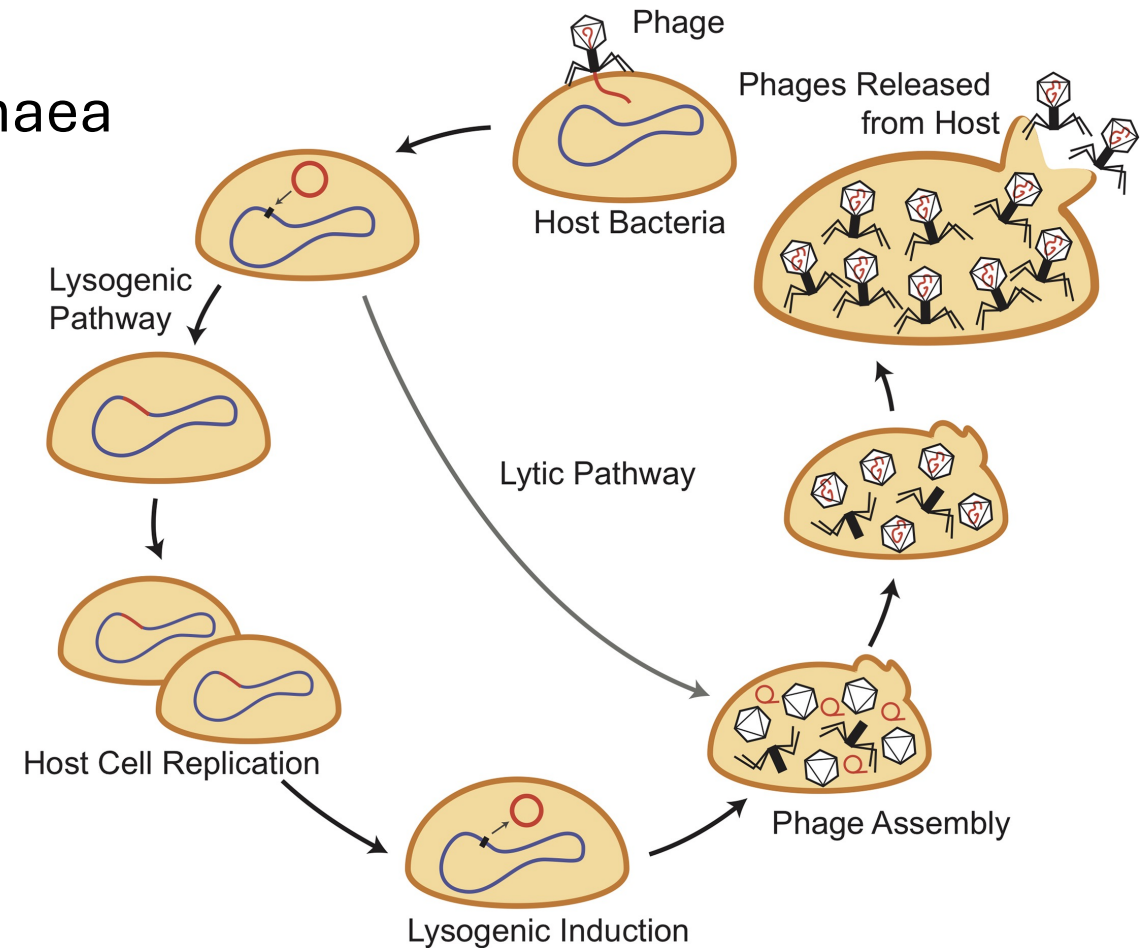
MBDP Metagenomics, 27.5.2026

Viral diversity is huge!



Virus-host interactions

- Viruses infect eukaryotes, bacteria, and archaea
 - control host abundance
 - may affect metabolism
(auxiliary metabolic genes, AMGs)
 - horizontal gene transfer
 - => indirectly affect global cycling
- Some viruses hijack other viruses



Viral genomes

Baltimore groups

Genetic material present in the virion

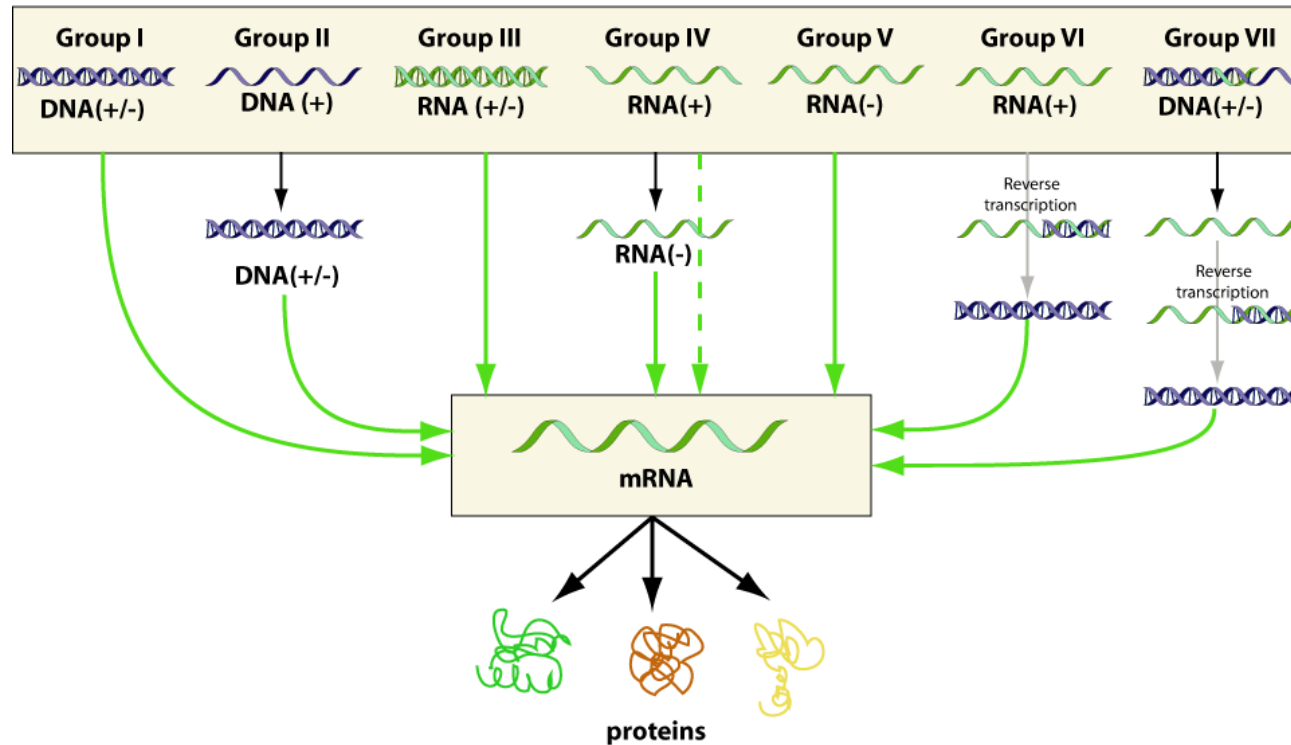


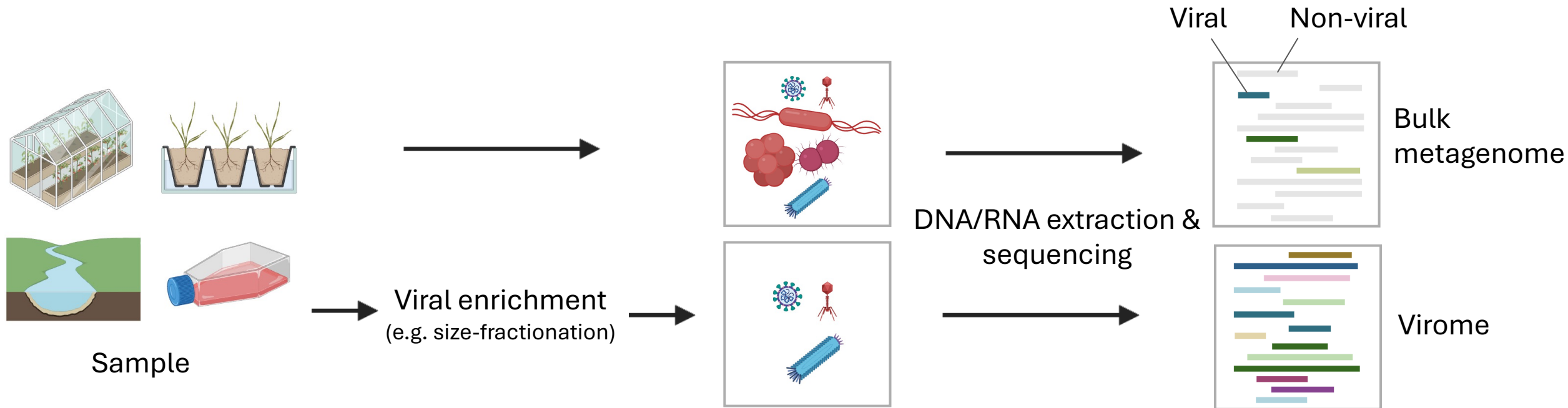
Image credit: ViralZone (SIB Swiss Institute of Bioinformatics),
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Viromics

Metagenomics = study of all genomes (total DNA) in an environment

Metatranscriptomics = study of all transcripts/other RNA elements (total RNA)

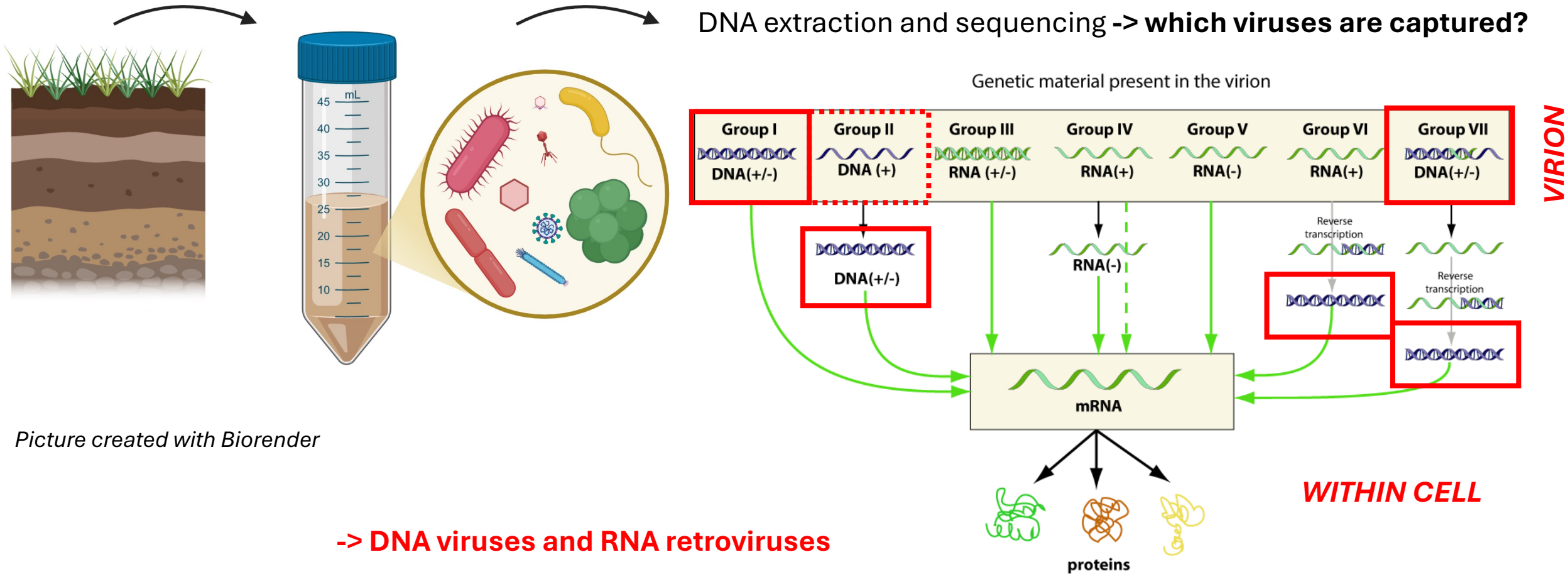
Viromics = study of all **viral genomes** (can be DNA or RNA)



Things to keep in mind when sequencing viruses

- What is being sequenced?
- All DNA, all RNA?
- Are library preparation protocols suitable for
 - dsDNA and/or ssDNA?
 - dsRNA and/or ssRNA?
- Size-selected viral fraction or bulk DNA/RNA?

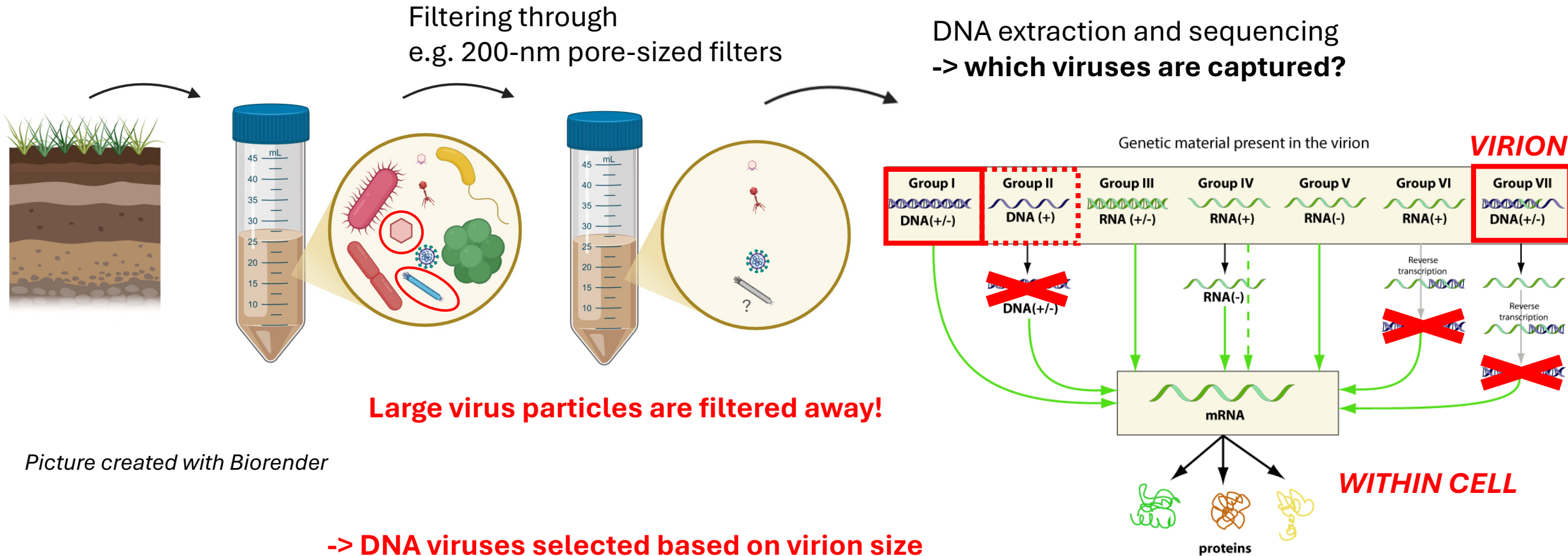
Example 1: bulk soil metagenome



Picture created with Biorender

Image modified from ViralZone (SIB Swiss Institute of Bioinformatics),
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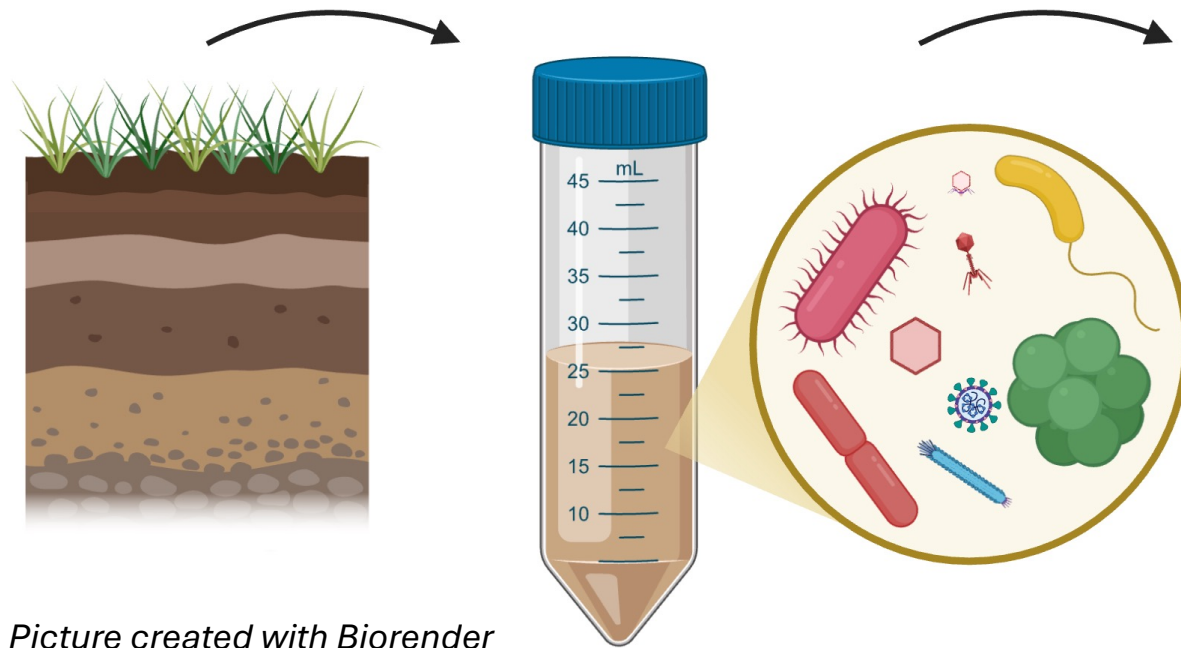
Example 2: soil size-fractionated virome (metaG)



Picture created with Biorender

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Example 3: bulk soil metatranscriptome



**-> RNA viruses and active DNA viruses
+ subviral RNA agents that don't produce particles**

RNA extraction and sequencing
-> which viruses are captured?

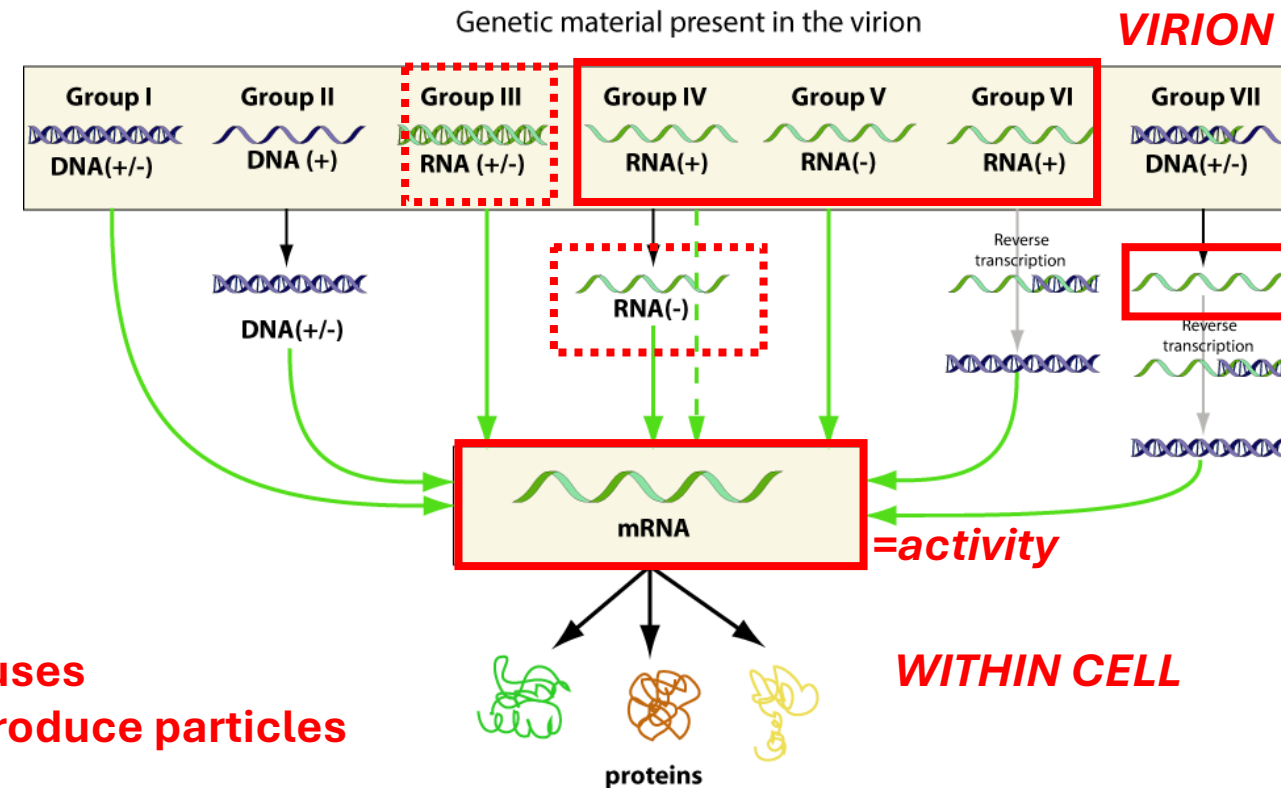
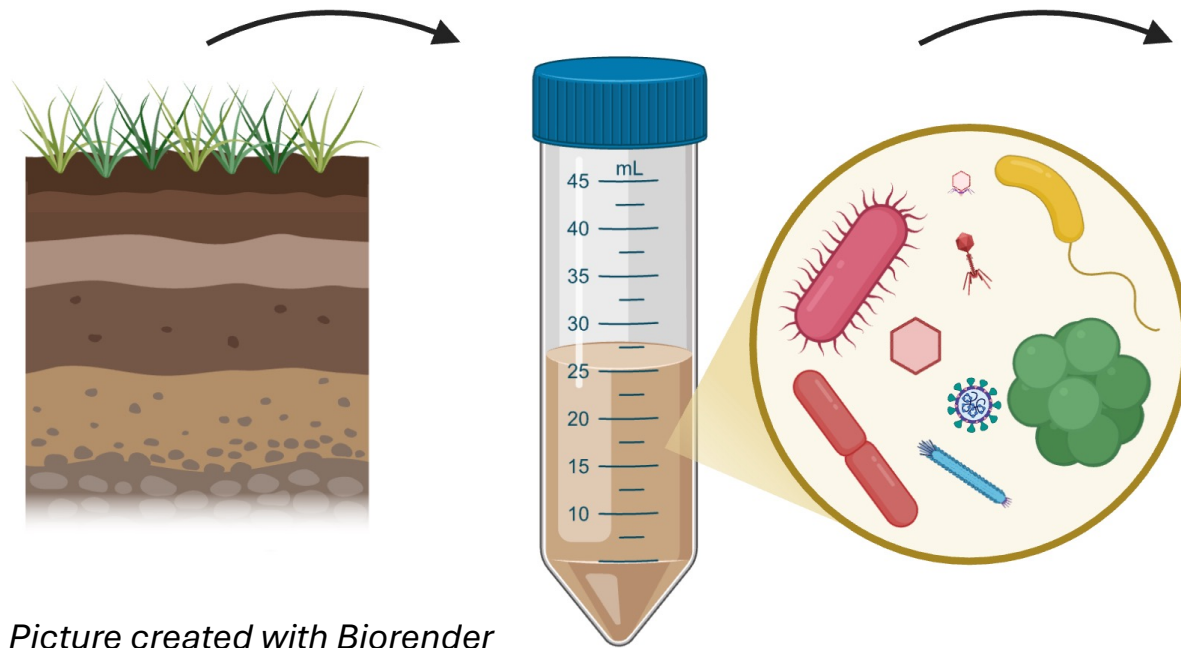


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Example 4: soil size-fractionated virome (metaT)



-> RNA viruses from size-selected virions

RNA extraction and sequencing
-> **which viruses are captured?**

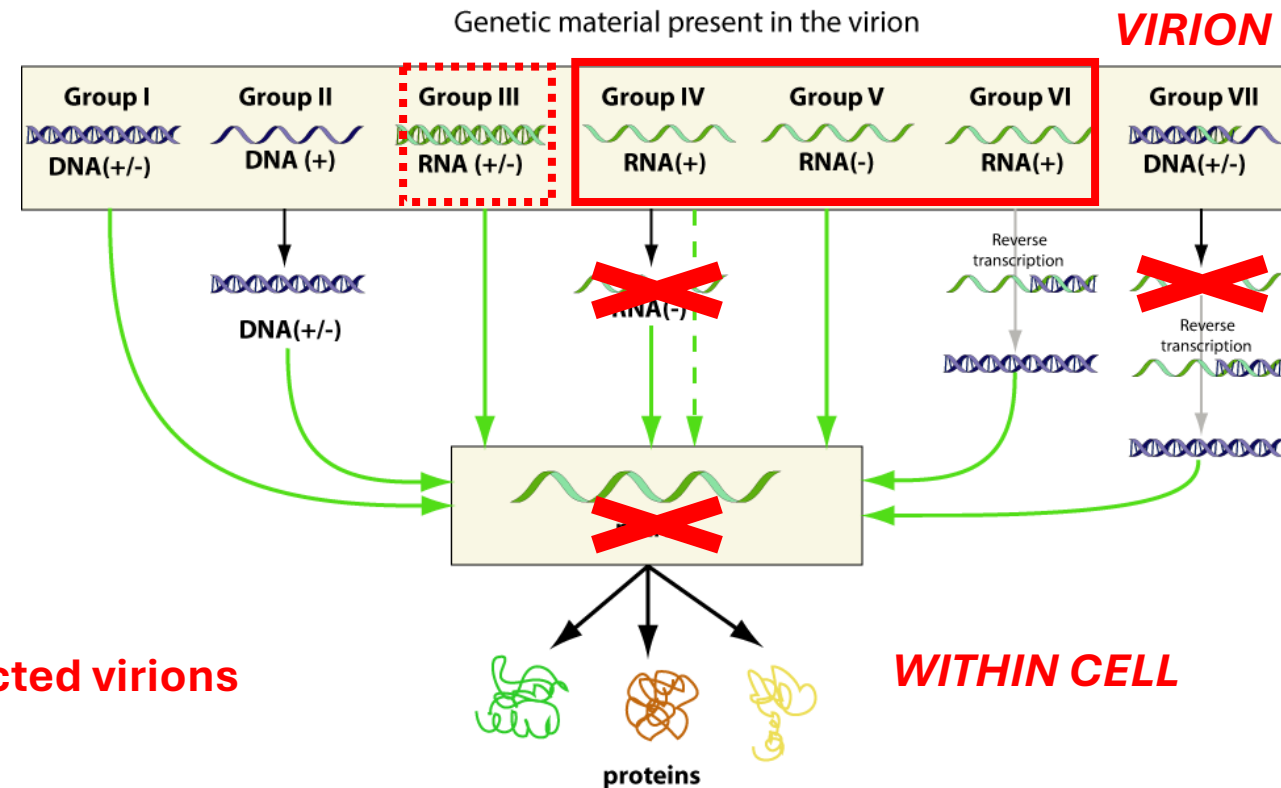


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Size-fractionated viromes vs bulk datasets

- In soil, size-fractionated viromes better represent viral communities (from virions)
- Bulk datasets enable capturing e.g. viruses that don't produce viral particles

> [Microbiome](#). 2024 Oct 7;12(1):195. doi: 10.1186/s40168-024-01905-x.

Viromes vs. mixed community metagenomes: choice of method dictates interpretation of viral community ecology

James C Kosmopoulos^{1 2}, Katherine M Klier^{1 3}, Marguerite V Langwig^{1 3}, Patricia Q Tran¹, Karthik Anantharaman^{4 5 6}

“We suggest that the choice of whether to target a metagenome or virome to study viral communities should be dependent on the environmental context and ecological questions being asked. However, our overall recommendation ... is to pair both approaches to maximize their respective benefits” (Kosmopoulos et al. 2024)

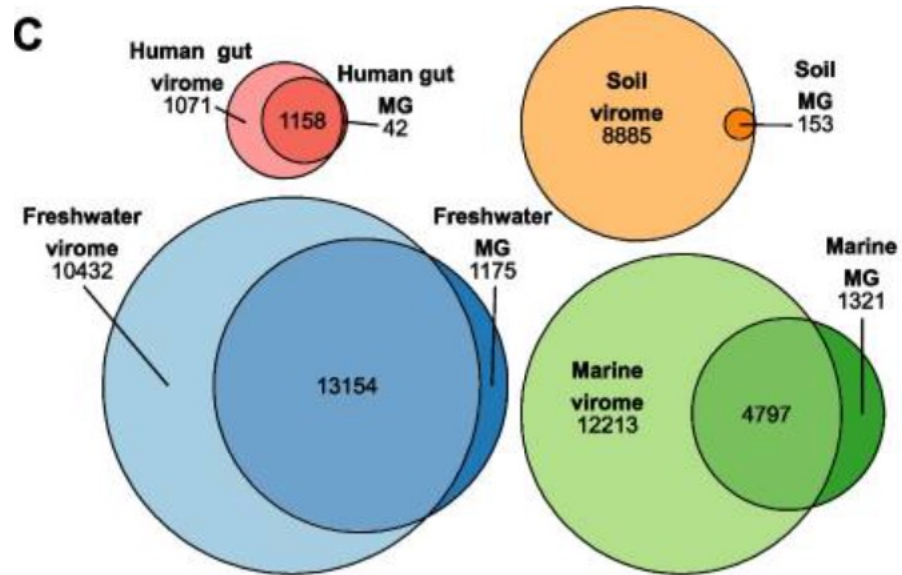


Image credit: Kosmopoulos et al., *Microbiome*, 2024,
[doi:10.1186/s40168-024-01905-x](https://doi.org/10.1186/s40168-024-01905-x)

Basic concepts (terms) in viromics

UViG = uncultivated virus genome

vOTU = viral operational taxonomic unit = species-rank group

Standard thresholds: 95% average nucleotide identity over 85% alignment fraction relative to the shorter sequence (suggested in *Roux et al 2019*)

vMAG = binned viral genome (from binning multiple contigs)

Perspective | [Open access](#) | Published: 17 December 2018

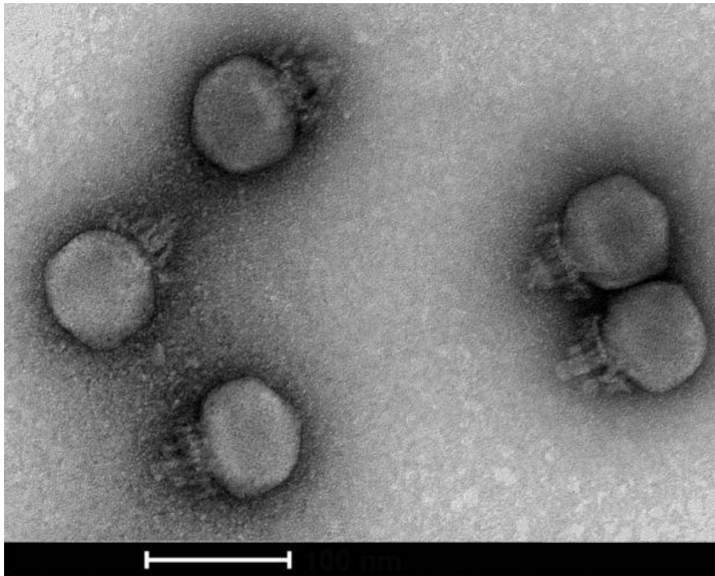
Minimum Information about an Uncultivated Virus Genome (MIUViG)

[Simon Roux](#) , [Evelien M Adriaenssens](#), [Bas E Dutilh](#), [Eugene V Koonin](#), [Andrew M Kropinski](#), [Mart Krupovic](#), [Jens H Kuhn](#), [Rob Lavigne](#), [J Rodney Brister](#), [Arvind Varsani](#), [Clara Amid](#), [Ramy K Aziz](#), [Seth R Bordenstein](#), [Peer Bork](#), [Mya Breitbart](#), [Guy R Cochrane](#), [Rebecca A Daly](#), [Christelle Desnues](#), [Melissa B Duhaime](#), [Joanne B Emerson](#), [François Enault](#), [Jed A Fuhrman](#), [Pascal Hingamp](#), [Philip Hugenholtz](#),
... [Emiley A Elie-Fadrosh](#)  [+ Show authors](#)

[Nature Biotechnology](#) **37**, 29–37 (2019) | [Cite this article](#)

Case study: crAssphage

- Discovered through metagenomics
(crass, **cross-assembly** software) (*Dutilh et al 2014*):
 - found in ~50% of individuals
 - represents up to 90% of human gut viromes
- Isolated afterwards (*Shkoporov et al 2018*):
 - predicted host and virion morphotype confirmed



Φ crAss001 virions, scale bar 100 nm

Image credit: Shkoporov et al., *Nature Communications*, 2018,

[doi:10.1038/s41467-018-07225-7](https://doi.org/10.1038/s41467-018-07225-7)

Case study: Megaphages

- Megaphages = phages with genomes of ≥ 500 kbp in length
- Largest reported phage genome so far: 841 kbp (*Chen 2025*)
- Metagenomically detected across diverse ecosystems
- Prevalent in human and animal gut (*Chen et al 2025*)
- Only a few cultivated
- Enlarged repertoire of genes: e.g., may encode ribosome proteins, AMGs, CRISPR-Cas systems...

► [Res Sq](#) [Preprint]. 2025 Aug 19:rs.3.rs-7356405. [Version 1] doi: [10.21203/rs.3.rs-7356405/v1](https://doi.org/10.21203/rs.3.rs-7356405/v1) [↗](#)

A prevalent huge phage clade in human and animal gut microbiomes

[LinXing Chen](#)^{1,*}, [Antonio Pedro Camargo](#)^{2,3}, [Yiting Qin](#)¹, [Eugene V Koonin](#)⁴, [Haoyu Wang](#)^{5,6}, [Yuanqiang Zou](#)⁵, [Yi Duan](#)^{7,8}, [Hao Li](#)¹

Case study: possible missannotations

Article | [Open access](#) | Published: 29 December 2022

Metagenomic analysis reveals unexplored diversity of archaeal virome in the human gut

[Ran Li](#), [Yongming Wang](#), [Han Hu](#), [Yan Tan](#) & [Yingfei Ma](#) 

[Nature Communications](#) **13**, Article number: 7978 (2022) | [Cite this article](#)

=> Human Gut Archaeal Virome Database (HGAVD)
>13-fold increase in archaeal virus diversity from human gut compared to previous studies

Matters Arising | [Open access](#) | Published: 17 July 2024

Inaccurate viral prediction leads to overestimated diversity of the archaeal virome in the human gut

[Cynthia M. Chibani](#), [Shiraz A. Shah](#), [Ruth A. Schmitz](#) & [Stephen Nayfach](#) 

[Nature Communications](#) **15**, Article number: 5976 (2024) | [Cite this article](#)

=> 72–83% of HGAD sequences classified as **non-viral** by six different viral prediction tools

Matters Arising | [Open access](#) | Published: 17 July 2024

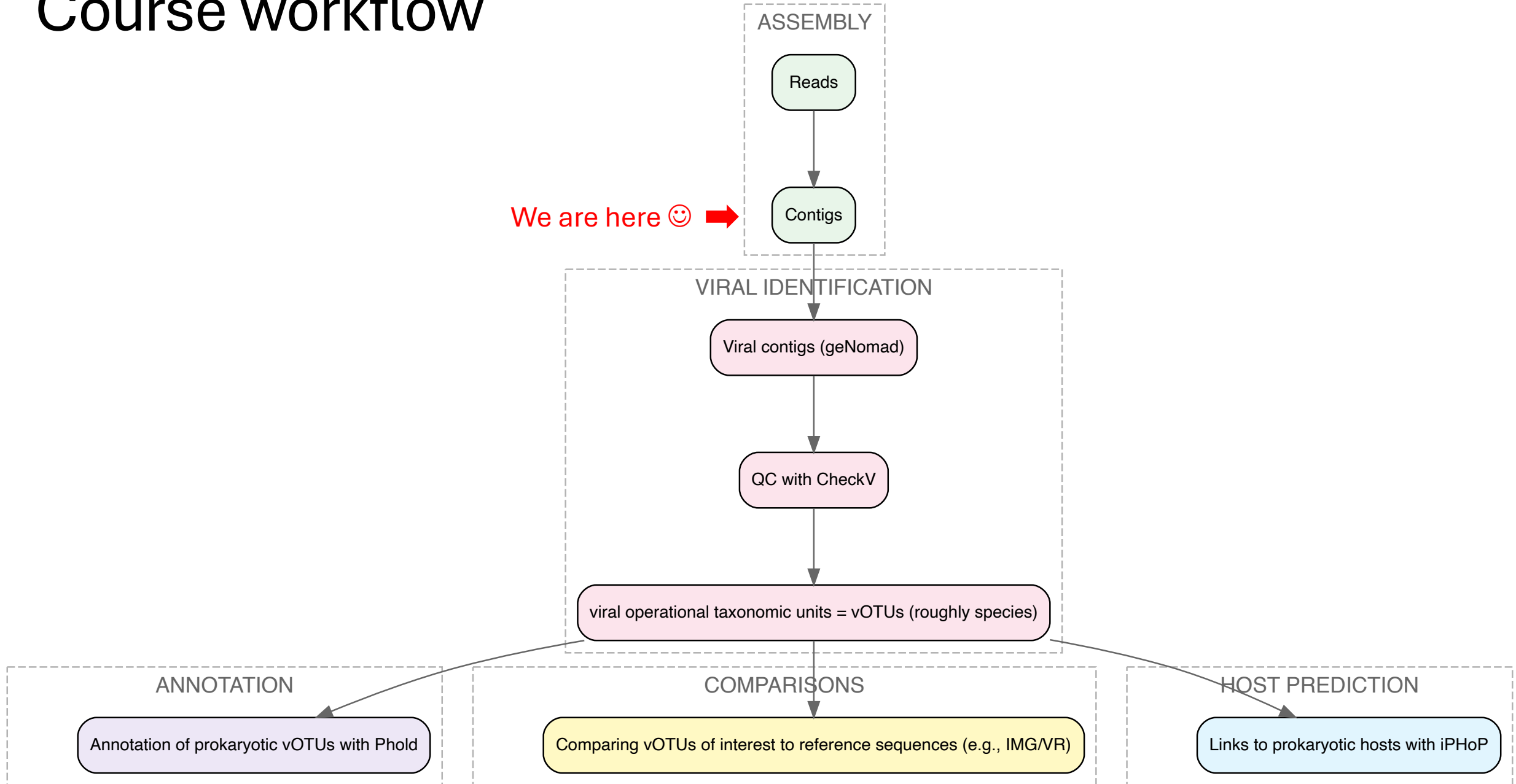
Reply to: Inaccurate viral prediction leads to overestimated diversity of the archaeal virome in the human gut

[Yongming Wang](#), [Ran Li](#) & [Yingfei Ma](#) 

[Nature Communications](#) **15**, Article number: 5977 (2024) | [Cite this article](#)

=> “We acknowledge that the original (published) HGAVD database likely contained high host sequence contamination rates based on re-analysis. For database profiling purposes, we advise using only sequences classified as ‘Complete Viruses’ and ‘High Confidence Viruses’ from the HGAVD to ensure accuracy and reliability.”

Course workflow



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